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MANAGEMENT SCIENCE
Vol. 70, No. 6, June 2024, pp. 4105–4114
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Transient Customer Response to Data Breaches of Their Information

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Received: April 20, 2021
Revised: May 25, 2022; June 3, 2023
Accepted: July 3, 2023
Published Online in Articles in Advance: January 31, 2024

https://doi.org/10.1287/mnsc.2021.01335

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Abstract. Cybersecurity breaches pose a substantial concern in the digital era. We investigate how customers respond to multiple unexpected data breaches of their information in India. Difference-in-differences estimates show that digital payments declined by 9% relative to cash payments immediately after an unexpected data breach in a food delivery platform, but the gap disappeared three months later. Customer entry and exit also exhibit weak, short-lived changes. Additional analyses on bank and online grocery data breaches uncover even weaker effects of data breaches. Our findings imply that the perceived benefits of convenience outweigh the costs of payment security risks.

History: Accepted by David Sraer, finance.

Funding: This work was supported by the MOE (Ministry of Education in China) Project of Humanities and Social Sciences [Project 20YJC790183], National University of Singapore [Start-Up Grant A-0003870-00-00], the National Natural Science Foundation of China [Grant 72150004], and the National Nature Science Foundation of China [Grant 72203112].

Supplemental Material: The online appendices and data are available at https://doi.org/10.1287/mnsc.2021.01335.

Keywords: cybersecurity breach • unexpected data breach • digital payment • cash payment • convenience • payment security • digital economy • India

1. Introduction

Digital services, such as food delivery and ride sharing, store and harvest private or even confidential consumer data. Given the massive volume of Internet activity, cybersecurity breaches can lead to large-scale data leakages and pose a substantial concern for users of digital services. Notable recent data breaches include Yahoo, Equifax, and eBay that affected 3 billion, 147 million, and 143 million accounts, respectively. It remains largely unknown how consumers respond to data breaches of their information. Do a breached digital service lose customers? How persistent is the effect? Does the data breach shift the composition of customers? We analyze consumer behaviors in an unexpected data breach to shed light on these open questions.

The data breach involved a leading online food delivery platform in India. The platform aggregates information on restaurants and allows consumers to order food and pay for the food delivery services by cash or digital payments (digital wallets and bank cards). The data breach was first reported by media on May 18, 2017, and involved 17 million users' names, email addresses, and passwords; no payment information was leaked according to the firm. We use the customer receipt-level transaction data from this platform to analyze how the data breach affected consumer behaviors. The payment mode information allows us to examine the relative change between cash and digital payments after controlling for the time trends in a difference-in-differences (DID) framework. Digital payments are expected to decline relative to cash payments if consumers are concerned about their payment security.

Following the data breach, the overall use exhibited a modest and temporary decline. Such a decline in overall use was driven by digital payments; cash payments increased despite the data breach. Digital payments declined by 9% relative to cash payments during the first month after the breach. This divergence reveals that users worried about payment security even when no payment information was leaked. However, the decline was transient. The difference between digital and cash payments vanished in the third month after the breach.

To gauge whether the data breach changes customer composition, we also examine customer entries and exits. Both entry and exit intensities changed little following the data breach. The entrants after the data breach do not differ, in terms of observable characteristics, from the entrants before the data breach. Interestingly, compared

with existing users who primarily used cash, those who primarily used digital payments were less likely to quit using the app following the data breach. As the prebreach usage of digital payments partly reflects demand for convenience, this result suggests that consumers who value convenience continue using the app but may switch from digital payments to cash payment, which is perceived to be safer.

Taken together, our analysis shows that the data breach indeed affects consumers in the short term and leads to a switch from digital payments to cash payments. In the long term, usages return to prebreach levels. Compared with surveys that directly elicit consumer perceptions of data breaches, our approach of analyzing consumer activity uses data with little measurement error and permits a more powerful test. A limitation is that we are unable to directly measure consumer perceptions from actual behaviors. We discuss several potential mechanisms through which the data breach has a transient effect on consumer behaviors. Our preferred interpretation is as follows: Although concerns for payment security triggered by the privacy leakage lead to a decline in digital payments in the short term, the perceived benefits of convenience outweigh the costs of the breach in the long term; as a result, uses return to prebreach level.

We also analyze the consumer response to two other unexpected data breaches, one on an online grocery store and the other one a bank, using administrative data from the breached firms. We find that these two data breaches have little effect on consumer behaviors. The stability of the results across the multiple breaches demonstrates that the observed transient response is not driven by app- or industry-specific idiosyncratic characteristics. Overall, the examined breaches have different causes and types of what is typically considered nonsensitive data involved. Similar to the data breaches we analyze, the majority of data breaches do not compromise financial or personally identifying information.[1] The transient impact of data breaches on consumer behaviors that we document in our analysis is conceptually applicable to other data breaches of nonsensitive data.

We contribute to an emerging literature on consumer response to data breaches. When some studies use survey data (Greene and Stavins 2017), evidence based on transaction data starts to emerge. For example, Schuh and Stavins (2016) and Mikhed and Vogan (2018) study how data breaches of bank account affect consumers' card use. Janakiraman et al. (2018) study how data breaches of a retailer affect their customers purchase from of the retailer. Our paper differs from the literature as the data breach of the food delivery platform only leaked platform login information and *did not* involve digital payment account information. Our data enable us to separately analyze the effects on platform use and digital payment adoption.

We also contribute to the empirical literature on the digital economy and private information sharing (Tang 2019, Argente et al. 2022). Most existing studies discuss convenience and private information security separately (Teo et al. 2015, Gu et al. 2017, Rysman and Schuh 2017). A few exceptions including Pentina et al. (2016) and Wottrich et al. (2018) use survey-based or experiment data and find that the perceived information security concern does not prevent the adoption or future use of private-information sensitive apps. Our paper shows that the digital payment declines in the short run but gradually recovers as time goes on, which support the consumers' tradeoff between payment security and convenience.[2]

Our findings shed light on the welfare implications of data ownership. Recent theoretical studies highlight nonrivalry and increasing returns to scale of data sharing; as a result, even in the presence of privacy considerations, consumer data sharing leads to social gains (Jones and Tonetti 2020) and contributes to economic growth through research and development (R&D) and knowledge accumulation (Cong et al. 2021). By indirectly showing that consumers are willing to share data, which can expose them to information security risk, in exchange for convenience, our analyses empirically corroborate these theoretical predictions.

2. Empirical Setting and Data Sources

We study how consumers respond to information leakage in an unexpected data breach. The breach occurred on a leading food delivery platform in India. The firm aggregates information on restaurants on an online platform and provides food ordering and delivery services from partner restaurants in select cities. On May 17, 2017, customer account information on the platform was discovered to have been sold on a dark web marketplace. On May 18, 2017, the firm confirmed this data breach, which was widely reported thereafter. The data breach involved 17 million users' names, email addresses, and passwords. The firm claimed that it had reset passwords for all affected users and logged them out of its app and website, and no payment information was leaked.

We use administrative data from the breached platform for our analysis. The data set contains microlevel information about food ordering and delivery from April 2016 to December 2017, in Delhi NCR, Mumbai, Bangalore, and Kolkata. Each customer account on this food ordering platform has a unique user ID, and we observe the city, subzone, and delivery time of every order made by each account.[3] For each order, we observe the name, unit price, and quantity of each food item. More importantly, we observe payment methods. Customers can pay by cash on delivery, digital wallet, credit card, or Internet bank. We consider all noncash payments as digital payments.

Compared with prior studies on consumer responses to data breaches that use survey data, we perform a more powerful test based on high-frequency transaction-based

spending from administrative data directly from the breached firm that have little measurement error.

3. Consumer Responses to the Data Breach

3.1. Change in Payment Modes

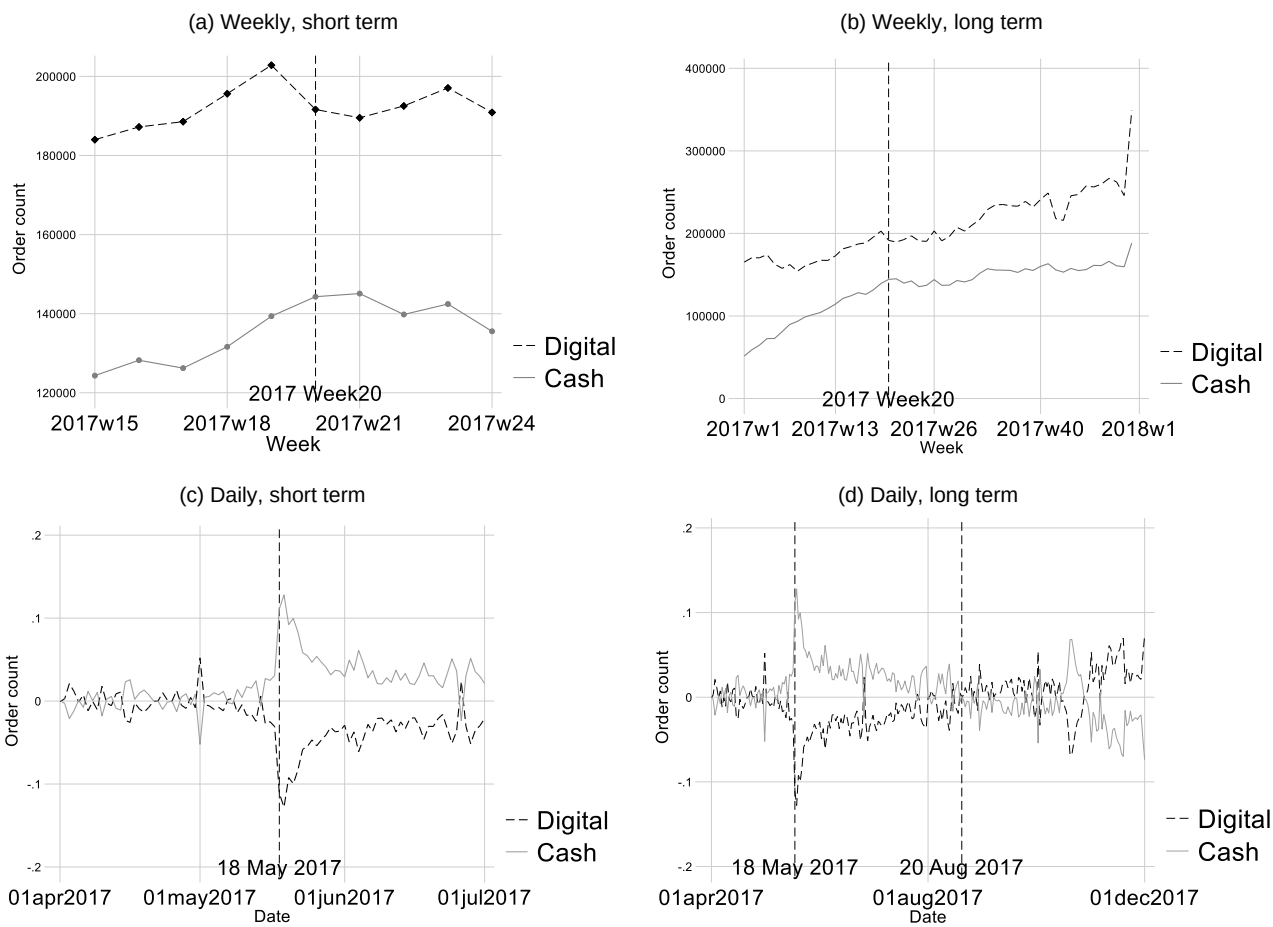
We begin our analysis with an unconditional comparison of the two payment modes. The breach was confirmed on the Thursday of week 20 in 2017 (May 18). We drop observations before April 1, 2017, to eliminate the mechanical effect of the November 2016 demonetization on cash availability.[4] We plot the total number of orders paid by digital and cash payments over time in Figure 1. Figure 1(a) shows that in the week of the breach, the number of digital payment orders experienced a sharp decline, whereas the number of cash orders increased. This divergence indicates that consumers switched from digital payments to cash payments after the breach. Figure 1(b) shows the relative decline in digital payment use last only temporarily. Digital payment usage gradually recovered, and the gap between digital and cash payments also recovered to the prebreach level around week 31. We then look into the daily time series seasonally adjusted as described in the online appendix, Section A.1.1. Figure 1(c) shows that the gap between the two lines substantially widened on May 18. As time went on, the gap gradually shrank to a negligible level (around August 20, 2017) as shown in Figure 1(d).[5] We examine the number of consumers and the rupee amount of sales for robustness and find similar patterns.

We then rigorously investigate how this exogenous shock changed consumers' payment choices. We construct a balanced panel data set of order characteristics for each payment mode in each week in each subzone and estimate the following DID panel regression:

$$\ln(1 + Y_{ptz}) = \mu_p + \lambda_{tz} + \beta \cdot (Treated_p \times T_t) + \varepsilon_{ptz}, \quad (1)$$

where p, t, and z represent payment modes (digital or cash), weeks, and subzones, respectively. Y_{ptz} is the number of orders paid by payment mode p in week t in subzone z.[6] We also consider daily frequency in the robustness tests. $Treat_p$ equals one if p is digital payment and equals zero if p is cash payment. T_t equals one if

Figure 1. Changes in Payment Modes: Order Count



Notes. The figure shows the time series of the number of orders paid by digital and cash payments, respectively. Dashed lines represent digital payments, and solid lines represent cash payments. The weekly sample is presented in (a) and (b). The seasonally adjusted daily sample mentioned in Section 3.1 is presented in (c) and (d).

week t is equal to or later than week 20 of 2017 in which the breach was announced and equals zero otherwise. Payment mode fixed effects μ_p control for fixed differences between digital and cash payments and subsume the treatment indicator. Week-subzone fixed effects λ_{tz} control for time trends and allow for heterogeneous time-varying shocks across different subzones. Robust standard errors are clustered at the city level.[7] By this specification, we eliminate the effects of any factors that may affect consumers' willingness to adopt the platform and focus on the difference between the adoption of digital payments and cash payments.

We apply Equation (1) to the payment mode-week-subzone panel data set. First, we restrict the sample to the period from week 16 to week 23, that is, four weeks before and four weeks since week 20. Column 1 of Panel A in Table 1 displays the results. The subzone-level number of digital payment orders declined by 9%, consistent with the short-term decline in Figure 1. We then use observations from week 16 to week 19 (the month before the breach) and from week 28 to week 31 (the third month after the breach) in column 2 of Panel A in Table 1. The statistically insignificant coefficient shows that digital payment uses were no longer lower than cash payment adoptions. We find similar patterns in the daily sample in the online appendix, Table A.1. Compared with cash payments, digital payments decreased in the month after the breach event but recovered in the third month after the breach.

In Panel B of Table 1, we return to the weekly sample and use sales and the price of order (i.e., the value of each payment) as the dependent variable, respectively. In the short run (columns 1 and 2), sales decline by a similar extent to order count, whereas the price of order does not change. In the third month (columns 3 and 4), there is little change.

We also analyze using alternative specifications. We exploit a user level sample in the online appendix, Section A.2.2, and find that consumers were less likely to choose digital payment in the short run after the breach. In the long run, the digital payment adoption recovered. Regarding the identifying assumption, Figure 1(a) shows

Table 1. Changes in Payment Modes

Panel A: Weekly

	Dependent variable: *Ln(1 + order count)*	
	(1) Post: the first month	(2) Post: the third month
Post × Digital	−0.090***	0.019
	(−6.32)	(0.75)
Payment mode fixed effects (digital dummy)	Yes	Yes
Week × subzone fixed effects	Yes	Yes
Observations	10,224	10,224
Observations with dependent variable = 0	944	730
R^2	0.976	0.975
Mean dependent variable	4.081	4.156
P value for wild *t*	0.058	0.529

Panel B: Alternative dependent variables

	Dependent variable			
	(1) *Ln(1 + sales)*	(2) *Ln(1 + order price)*	(3) *Ln(1 + sales)*	(4) *Ln(1 + order price)*
	Post: The first month		Post: The third month	
Post × Digital	−0.101**	−0.004	0.092	0.020
	(−3.51)	(−0.31)	(1.46)	(1.37)
Payment mode fixed effects (digital dummy)	Yes	Yes	Yes	Yes
Week × subzone fixed effects	Yes	Yes	Yes	Yes
Observations	10,224	9,090	10,096	9,332
Observations with dependent variable = 0	944	0	730	0
R^2	0.952	0.817	0.951	0.815
Mean dependent variable	9.455	5.977	9.638	5.960
P value for wild *t*	0.064	0.850	0.056	0.166

Notes. Subzone-payment mode-weekly (daily) level regressions estimating the response of payment modes to the data breach of the food delivery platform. In Panel A, we compare the month before the breach with the first and the third month after the breach respectively, and the dependent variable is the natural logarithm of the number of orders. In Panel B, we use the two sample periods respectively and replace the dependent variable with sales and the price of orders, respectively. The mean dependent variable, $\ln(1 + y)$, is reported at the bottom. Robust standard errors are clustered at the city level. The t statistics are reported in parentheses. For significance testing, the $t(G - 1)$ distribution is used to obtain p values for cluster-robust t statistics where G is the number of clusters.

*, **, and ***Significance at the 10%, 5%, and 1% levels, respectively.

parallel prebreach trends in unconditional patterns. We also conduct rigorous regression analysis in the online appendix, Section A.2.3. In Figure A.3, the coefficients for prebreach weeks are all statistically and economically indistinguishable from zero, and there is a sharp decline in digital payments immediately after the breach.

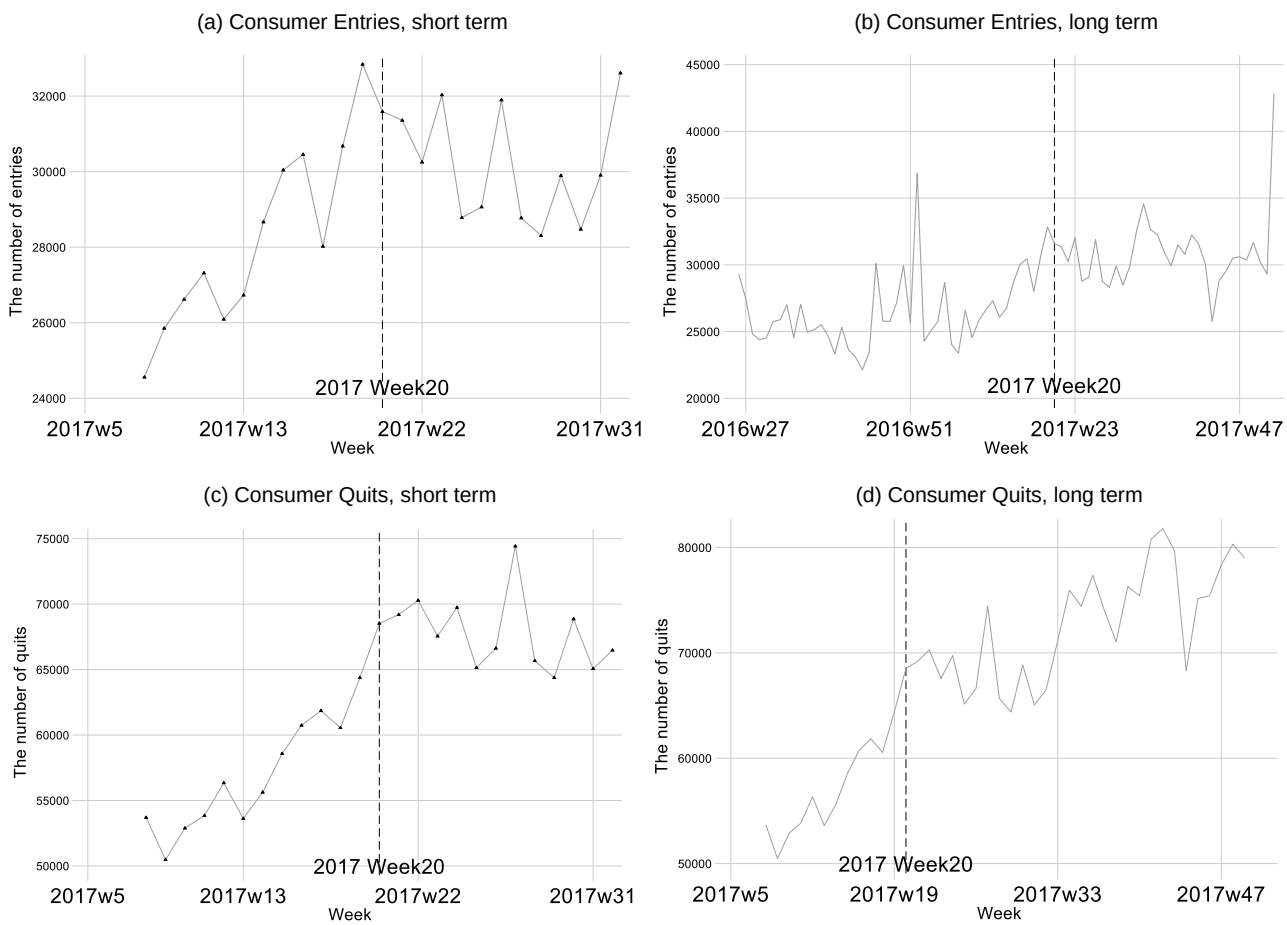
3.2. Customer Entries and Quits

Next, we analyze customer entries into the platform. Entrants presumably know about the data breach that happened on the app, given the wide media coverage of the data breach. They did not have their data already on the app. They faced a tradeoff between sharing their data to the app and hence being exposed to breaches on the app versus the utility from using the app. Figure 2(a) plots the weekly time series of the number of new customers in the short term. The time series had an upward trend before week 20. The trend was reversed and went down until week 22, which indicates that the data breach discouraged consumers' adoption of this platform. However, the decline only persisted for three weeks, and its impact is small in the long run as shown in Figure 2(b).

We apply our main DID specification to examine entry to the app. For every week and each subzone, we calculate the number of digital payment orders and cash orders by entrants. As can been seen from the online appendix, Table A.3, the number of digital payment orders by entrants decrease 4.3% per week in the two weeks after the data breach relative to the change for the number of cash orders by entrants. However, the difference becomes insignificant in the third month after the data breach. The regression results which are obtained with tight fixed effects confirm the unconditional patterns of entry: Digital payments by entrants experiences a dip immediately after the data breach relative to cash payments but rebounds soon.

We also explore how the breach affected the composition of customers entering the platform. In Table 2, we compare the characteristics of the first order made by customers who entered one month before (i.e., between April 18 and May 17) and after the breach (i.e., between May 18 and June 17). *Quantity* is the average number of

Figure 2. Consumer Entries and Quits



Notes. The figure shows the time series of consumer entries and quits. Entries that are defined as the first appearance of the user in the sample are presented in (a) and (b). Quits are presented in (c) and (d). The week of a quit is defined as the first week of four-week inactivity.

Table 2. Data Breach and Customer Entries

Variable	Before	After	Diff/p value
Price	1,507.613	1,489.669	−17.944 [0.241]
Unit Price	181.840	183.839	1.999*** [0.000004]
Quantity	3.165	3.141	−0.024* [0.069]
Vegetarian Index	0.360	0.357	−0.003* [0.099]
Observations	130,738	135,600	266,338

Note. This table shows the t tests comparing the means of the characteristics of the first order by the customers who entered one month before and after the data breach.

items in an order. It measures the average size of orders by the user. *Unit Price* is the average unit price of the food items in the orders by the user and acts as a proxy for income level. *Price* is the price of the entire order. *Vegetarian Index* is the ratio between the number of orders without meat and the number of orders with meat or fish. We infer whether an order includes meat or fish by the names of the food items in the order. We use the *Vegetarian Index* to infer the eating habits and investigate composition change of customers. The differences in *Price* and *Vegetarian Index* are statistically insignificant. The differences in *Unit Price* and *Quantity* are statistically significant at the 1% and 10% level, respectively, but the magnitude of the differences is negligible: the change in *Unit Price* (*Quantity*) accounts for only 1% (0.8%) of the mean of customers who entered before the breach. Overall, there are little changes in the characteristics of new users' ordering behaviors, suggesting that the composition of new customers likely remains stable after the breach.[8]

We then investigate consumer exits (quits) from the platform. We define quit as a four-week (or longer) inactivity.[9] Specifically, if a user has no orders from week t to week $t+3$, we consider the user quits in week t. We compute the weekly number of quits and plot the time series in Figure 2(c). There was an upward trend in week 20, but that trend could be due to the momentum of the upward trend that started in week 18. Besides, the trend was persistent only for a couple of weeks, and its magnitude is minor in the long run as displayed in Figure 2(d). The evidence on entries and quits suggests that the data breach had little impact on consumers' adoption of the platform.

3.3. Individual Consumer Level Analysis of Quits and Payment Changes

Motivated by the differential reaction between cash and digital payment to the data breach, we explore whether existing consumers change their behaviors after the data breach. Their data on the app were leaked; the possibility that an unauthorized third party may exploit the breached information and bring losses to customers raises concerns for payment security. If they continue to use the app and stick to digital payments after the breach, their activity generates more payment data on the app (such as saved card numbers and reset passwords). For the new data being generated, leakage risks have not been realized but consumer perception of the leakage risks might change because of the breach.

To analyze quits at the individual consumer level, we restrict the analysis to the sample of prebreach active users, that is, those who order at least once per week in the four weeks before May 18.[10] We construct a cross-sectional sample for these active users with a dummy *Quit* that equals one if the consumer quits within four weeks after week 20 and equals zero otherwise. Because the dependent variable is a dummy, we apply the following logistic regression:

$$\Pr(Quit_i = 1) = \Phi\Big(\beta_0 + \beta_1 Digital_Ratio_i + \sum \gamma_j Control_{ij} + \lambda_c\Big), \tag{2}$$

where $\Phi(x) = \exp(x)/(1+\exp(x))$.

The key independent variable is *Digital_Ratio*, the ratio between the number of digital payments and the total number of payments from April 1 to May 17 of 2017. The variable measures the extent to which the consumer relies on digital payments. $\boldsymbol{\lambda_c}$ is city fixed effects. We include the following control variables: *Quantity, Unit price, Vegetarian Index, Account age,* and *Frequency. Account age* is the number of weeks between the first order by the user and week 18 of 2017. This variable is a proxy for how long the users had adopted the platform for. *Frequency* is the average number of orders per week by the user (during the period from his first order to May 18, 2017. The other three variables are constructed analogously as before.

In Table 3, Panel A shows the summary statistics, and column 1 of Panel B displays the estimation results. The average marginal effects are −0.103, which indicates a negative correlation between the odds of quits and *Digital_Ratio*. The results change little qualitatively when we include all controls in column 2.

We conduct an additional test to show that digital payment users switched to cash payments:

$$\Delta Digital_Ratio = \beta_0 + \beta_1 Digital_Ratio_i + \sum \gamma_j Control_{ij} + \lambda_c + \varepsilon_i, \tag{3}$$

We restrict the sample to "active" users identical to those in Equation (2) and drop users who quit in week 21 (who hence have no payment records after week 21). Then we compute the $\Delta Digital_Ratio$, which equals the *Digital_Ratio* during the period from week 21 to week 24 (i.e., the month after the breach) minus the *Digital_Ratio* used in Equation (2) (i.e., the ratio before the breach). The independent variables are similar to those in Equation (2).

Table 3. Quits and Payment Switches

Panel A: Summary statistics

Variable	Observations	Mean	Standard deviation
Quit	28,930	0.18	0.38
Digital_ratio	28,930	0.63	0.30
Quantity	28,930	2.71	1.18
Account Age	28,930	39.36	17.56
Frequency	28,930	1.60	1.19
Unit price	28,930	164.63	49.89
Vegetarian Index	28,930	0.18	0.15
ΔDigital_Ratio	27,117	0.01	0.26

Panel B: Regression results

	Dependent variable					
	Quit LOGIT coefficient		*ΔDigital_Ratio* OLS coefficient		*ΔLn(1 + Digital payments)* OLS coefficient	
	(1)	(2)	(3)	(4)	(5)	(6)
Digtal_ratio	−0.711***	−0.484***	−0.089***	−0.097***	−1.163***	−1.002***
	(−14.16)	(−18.46)	(−6.10)	(−6.65)	(−50.22)	(−40.17)
Quantity		−0.014		0.002		0.003
		(−0.93)		(1.53)		(1.53)
Account Age		−0.028***		0.002***		−0.027***
		(−18.04)		(11.34)		(−55.24)
Frequency		−0.378***		0.005*		−0.182***
		(−39.99)		(2.66)		(−11.13)
Unit price		−0.003***		0.000***		0.000*
		(−3.21)		(6.52)		(2.42)
Vegetarian index		−0.015		0.006		0.076
		(−0.30)		(0.58)		(1.24)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	28,930	28,930	27,117	27,117	27,117	27,117
R^2			0.013	0.024	0.137	0.372
Pseudo R^2	0.014	0.063				
Mean dependent variable	0.179	0.179	0.014	0.014	−1.641	−1.641
p value of wild *t*	0.093	0.027	0.086	0.084	0.011	0.019
Marginal effects of *Digital_Ratio*	−0.103***	−0.067***				
	(−14.38)	(−19.53)				

Notes. User-level regressions estimating the quits and payment switches of consumers after the data breach event. The sample period is cross-sectional and is constructed based on the period between the month before and the month after the breach event. Panel A displays the descriptive statistics of the sample, and Panel B displays the estimates. In columns 1 and 2, the dependent variable is an indicator that equals one if the consumer quit within four weeks after the breach. In columns 3 and 4, the dependent variable is the change in the digital payment ratio. In columns 5 and 6, the dependent variable is the natural logarithm of the digital payment. Robust standard errors are clustered at the city level. The *t* statistics are reported in parentheses. The mean dependent variable is reported at the bottom to assess marginal effects.

*, ** and ***Significance at the 10%, 5%, and 1% levels, respectively.

Columns 3 and 4 of Panel B in Table 3 display the results. The decline in the digital payment ratio was greater for consumers with higher digital payment ratios. In other words, they switched to cash to a larger extent. To eliminate the possibility that the greater decline is due to higher baseline of consumers with higher digital payment ratios, we replace the dependent variable with the natural logarithm of the digital payment count to measure growth rates. Columns 5 and 6 in Panel B of Table 3 report the estimates which indicate similar results.[11]

Columns 3 to 6 of Panel B in Table 3 indicate that digital payment users who prefer convenience more were reluctant to leave the platform, while cash users who prefer convenience less were more likely to quit the platform. Instead of immediately quitting, digital payment users sacrifice the convenience of digital payments to secure payment security but still retain the convenience of the takeout ordering platform.

4. Discussions of Potential Mechanisms for the Observed Consumer Responses

Compared with survey-based studies, our approach of analyzing consumer activity does not rely on participant self-report and recall and therefore is not subject to being

incomplete due to limits of memory, under-reported due to forgetting past experiences, or over-reported due to being influenced by media coverage. A limitation is that we are unable to directly measure consumer perceptions from actual behaviors. In this section, we discuss potential mechanisms through which the data breach affected consumer behaviors.

The data breach led to an immediate decrease in digital payments relative to cash payment despite that payment information was not leaked. One possibility is that initially the app users thought hackers could capture payment information, they switched to using cash payments until they realized the breach did not jeopardize their payment security.[12] Given that the initial discovery of the breach was limited to observers of the dark web marketplace and the company confirmed the data breach on the next day, most users learned about the breach from the platform. We believe that users were aware of the platform's claim that payment information was not leaked. It is possible that despite the platform's claim, users might be wary of payment security. Such concerns for payment security drive the initial response of digital payments.

The recovery of consumer demand to prebreach levels by the third month after the breach shows that the impact of the data breach is transient as opposed to permanent. We consider several explanations for this transient effect. First, we can assess the effect through the lens of a trade-off between convenience and payment security. When personal information is leaked in a data breach, consumers' decisions of changing their usage patterns depend on how much utility they gain from the convenience of using the app relative to the disutility associated with the breach. Observed behaviors allow us to indirectly infer consumer preferences: Although the concerns for payment security triggered by the privacy leakage lead to a switch away from digital payments in the short term, the perceived benefits of convenience outweigh the costs in the longer term; as result, uses return to prebreach levels. Admittedly, the costs can evolve over time due to either changes in the breach severity or consumer perception changes. Our analysis does not disentangle these different mechanisms. We leave this interesting area for future research.

Second, we consider the possibility that the breached firm engages in safety measures ex post. We searched news outlets and various Internet sources to track the firm's response to the breach. The firm contacted the hacker to delete the breached data and introduced monetary incentives for reporting security flaws in its existing bug bounty program. Such measures signal that the firm is increasing cybersecurity investments to reduce future breach risks and can potentially mitigate consumer concerns for future data breaches. In general, it is difficult for the increased cybersecurity investments to neutralize all future data breaches due to the enormity of potential system vulnerabilities, and as a result, repeated data breaches are common. In the online appendix, Section A.3.2, we further discuss repeated data breaches and the firm's mitigation measures and acknowledge that the measures by the breached firm may partially mitigate consumers' concerns about future data breaches. In addition, we find no evidence that the food delivery platform tried to retain its users by promotion using Google search index in the online appendix, Section A.2.6.

We also consider the role of inattention, that is, only a small fraction of the app users were aware of the data breach. At its face value, this seems unlikely given the large and significant drop in electronic payment immediately following the breach, and that the company had reset the passwords for all affected users and logged them out of the app and website. If, for some reasons, only a small fraction of the app users learned about the data breach initially, more app users should become aware of the data breach over time, leading to a persistent or even increasing reaction to the data breach. We do not see this in the data. Therefore, the only plausible form of inattention to explain our findings is that app users become inattentive over time (i.e., they forget). This explanation of limited memory is consistent with the trade-off interpretation: the affected consumers do not view the data breach as a big deal, otherwise they would not forget. We also exploit the geographic variation in attention as measured by Google search volume to test the role of inattention in the online appendix, Section A.3.1, and find no support for the inattention explanation.

Finally, the industrial organization structure of the food delivery industry may affect our findings. If there was only one major and reliable platform in a city, it could be mechanical that the consumers cannot migrate to other options. In this case, the limited options rather than the choice for convenience drive our results. This is unlikely to be the case as there were at least three major food delivery platforms, including the breached platform, in each of the four cities included in our sample. Therefore, consumers can migrate to another platform if they wish to.

5. Additional Analyses and Discussions

5.1. Other Data Breaches

We also analyze the consumer response to two other unexpected data breaches using administrative data from the breached firms. The two breaches involved a leading online grocery store and a leading national bank in India. They stem from different causes, but similar to the food delivery platform's breach and the majority of other data breaches, they involve what is typically considered nonsensitive data. In these two data breaches, we find weaker responses (online appendix, Sections A.4 and A.5). We also conduct additional robustness

checks by restricting the samples to the two overlapped cities Mumbai and Bangalore in Online Appendix E and find similar results. By studying multiple unexpected data breaches across different industries, we ensure that our results are not driven by app- or industry-specific idiosyncratic characteristics.

5.2. Statistical Inference

In our baseline results, we rely on Stata's default small-sample corrections to mitigate the potential downward bias of clustered standard errors when the number of clusters is low. We also use the wild cluster bootstrapping tests and show robustness of our statistical inference to computing two-way clustered standard errors in Online Appendix D. The largely null results raise another concern of whether the effect is close to zero or the data does not have enough statistical power. To address this concern, we implement the "serial-correlation-robust" (SCR) power calculations (Burlig et al. 2020), which allow for arbitrary (nonconstant) serial correlation in panel data applications, for our regression analyses in the online appendix, Section A.6.1. We show that our null results are not driven by insufficient statistical power. In our setting, a few factors boost the statistical power. First, the data were collected directly from the apps' transaction records and have very little (if any) measurement error. Second, we use weekly or daily data in our analysis, which provided high-frequency observations to detect the effect of data breaches.

5.3. External Validity

We draw our findings from multiple data breaches in India. For the following reasons, we believe that our findings can be applicable to other data breaches and other countries. First, similar to the data breaches that we analyze, the majority of data breaches do not involve leakage of sensitive personal information. Second, existing studies in household finance (D'Acunto et al. 2019) document that Indian consumers behave similarly to those in other countries. Third, a number of survey and experimental studies conducted in the United States and other countries show a paradox that consumers' self-reported preferences for sharing personal data are inconsistent with their actual behaviors. By analyzing consumer transactions, our study complements these survey and experimental studies to provide additional empirical estimates for consumers' personal data sharing preferences and how the preferences affect their data-sharing choices. In the online appendix, Section A.6.2, we provide additional details and discussions for external validity.

5.4. Implications for Data Protection Policies

In recent years, growing concerns for consumer privacy have prompted the enactment of data protection laws such as EU's General Data Protection Regulation. An important objective of these laws is to restrict collection of personal data, which can be misused for unwarranted reasons (e.g., discrimination, exploitation of consumer vulnerability) or leaked in cybersecurity breaches. These undesirable costs need to be weighed against the improved access to products and services and the substantial convenience enabled by big data technologies. In this study, we focus on the setting of data breaches, a risk factor from which data protection laws aim at protecting consumers, to analyze this tradeoff. By inferring consumer preferences from observed behaviors in unexpected data breaches, we empirically demonstrate that for consumers, the benefits of convenience outweigh the cost of concerns for payment security. Our findings are related to the theoretical framework of Liu et al. (2021) that models consumer preference for personal data sharing as a mechanism for concealing behavioral vulnerabilities. In their framework, restricting data sharing is more beneficial to consumers for temptation goods such as gambling or video games; for normal consumption goods such as food delivery, grocery, and banking service that we study in this paper, convenience dominates the concern for personal information security. In the online appendix, Section A.6.2, we provide additional discussions for the policy implications.

6. Conclusion

This paper investigates how consumers respond to unexpected data breaches of their personal information. We find that after an unexpected data breach, customers of a food delivery platform switched from digital payments to cash in the short run. In the long run, use of digital payments recovered. Customers' entry and exit intensities changed little following the data breach. Existing users who primarily used digital payments are less likely to quit the platform following the data breach, compared with those who primarily used cash. Additional analyses on bank and online grocery data breaches uncover even weaker effects. Overall, our analyses show that data breaches are indeed recognized by affected users in the short term. In the long run, usage returns to prebreach levels.

Our analyses use actual transaction data to contribute to our understanding of how data breaches affect consumer behaviors. Our findings imply that the perceived benefits of convenience outweigh the costs of payment security risks. The transient impact of data breaches on consumer behaviors is conceptually applicable to other data breaches of nonsensitive data. Consumers may respond more strongly to data breaches that compromise financial or personally identifying data.

Acknowledgments

The authors thank David Sraer (the editor), an anonymous associate editor, and four anonymous referees for comments. The authors thank Zefeng Chen, Yeqing Zhang, and participants at the Georgetown Global Virtual Seminar

Series on Fintech, Asian Meeting of the Econometric Society 2022, China Fintech Research Conference 2022, the 5th Big Data, AI and Fintech Conference, and Central University of Finance and Economics for helpful comments and discussions.

Endnotes

[1] The U.S. Government Accountability Office (2007) report on consumer data breaches considers "sensitive personal information" as financial information (bank account, credit card, financial investments, financial transactions) or personal identification information (Social Security numbers, government issued IDs). According to this definition, only 42 or 7.3% of the 577 data breaches that took place during the 11 years from 2011 to 2021 as compiled by Internet security service Have I Been Pwned (https://haveibeenpwned.com/) involved leakage of some sensitive personal information; the other 535 or 92.7% of the breaches did not involve leakage of any sensitive personal information.

[2] The reversal in the effects we find also differs from the findings in the data breach literature. Greene and Stavins (2017) examine consumer rating on security, but only explore payment use in the long run. Schuh and Stavins (2016) explore in the short run, Mikhed and Vogan (2018) find no change in card use, Janakiraman et al. (2018) find significant results for both short run and the long run. Kwon and Johnson (2015) find long run effects only.

[3] Subzone is a geographic unit in India and is smaller than a city. In our sample, each city has 159.5 subzones on average.

[4] Chodorow-Reich et al. (2020) show that, although the demonetization led to an immediate and geographically uneven cash shortage, the cash shortage was alleviated in the subsequent several months. By using a sample period starting in April 2017, more than four months after the demonetization, we by and large remove the impact of demonetization-induced cash shortage.

[5] There was a spike at the end of 2017 in Figure 1(b). There was a similar spike at the end of 2016, which indicates that the spike was a regular seasonal pattern.

[6] We add one to the Y to avoid missing value after taking logarithm. We apply this strategy for all regressions that take logarithm to the dependent variables. We also address zero-valued observations using inverse hyperbolic sine transformation and Poisson regression in Online Appendices B and C.

[7] As the number of cities is low, standard errors clustered at the city level may be downward biased. In our baseline results, we rely on Stata's default multiplicative small-sample correction to the asymptotic variance estimator as well as Stata's default degree of freedom adjustment in the significance testing. We also use the wild cluster bootstrapping tests to correct for the small number of clusters. We also apply the two-way clustering in Online Appendix D.

[8] We cannot fully reject a change in the customer composition as we do not observe demographic characteristics in our data. Prior research has documented that certain demographic groups, notably female and older people, are more sensitive to data privacy issues than others (Tang 2019, Prince and Wallsten 2022).

[9] As a comparison, users who ordered at least once in the fourth month before the breach on average placed one order for every 10.4 days during the following three months.

[10] May 18 (the date of breach announcement) was the Thursday of week 20. Even if a consumer wanted to quit immediately after the breach, she may still have had orders in week 20 because we use weekly sample here. Hence, the earliest week for quits is week 21.

[11] The value of *Digital_Ratio* is between zero and one, and the coefficient measure the marginal effect when *Digital_Ratio* increases from zero to one, and hence the big coefficients in columns 5 and 6 are not surprising.

[12] It is also possible that the app users may underestimate how damaging the data breach was and under-react initially. Under this alternative scenario, they should know better over time and correct their initial under-reaction. If this is the case, we would have observed persistent or even increasing reaction to the data breach. We do not see this in the data.

References

Argente D, Hsieh CT, Lee M (2022) The cost of privacy: Welfare effects of the disclosure of Covid-19 cases. *Rev. Econom. Statist.* 104(1):176–186.

Burlig F, Preonas L, Woerman M (2020) Panel data and experimental design. *J. Development Econom.* 144:102458.

Chodorow-Reich G, Gopinath G, Mishra P, Narayanan A (2020) Cash and the economy: Evidence from India's demonetization. *Quart. J. Econom.* 135(1):57–103.

Cong LW, Xie D, Zhang L (2021) Knowledge accumulation, privacy, and growth in a data economy. *Management Sci.* 67:6480–6492.

D'Acunto F, Prabhala N, Rossi AG (2019) The promises and pitfalls of robo-advising. *Rev. Financial Studi.* 32(5):1983–2020.

Greene C, Stavins J (2017) Did the Target data breach change consumer assessments of payment card security? *J. Payments Strategy Systems* 11(2):121–133.

Gu J, Xu YC, Xu H, Zhang C, Ling H (2017) Privacy concerns for mobile app download: An elaboration likelihood model perspective. *Decision Support Systems* (94):19–28.

Janakiraman R, Lim JH, Rishika R (2018) The effect of a data breach announcement on customer behavior: Evidence from a multichannel retailer. *J. Marketing* 82(2):85–105.

Jones CI, Tonetti C (2020) Nonrivalry and the economics of data. *Amer. Econom. Rev.* 110(9):2819–2858.

Kwon J, Johnson ME (2015) The market effect of healthcare security: Do patients care about data breaches? Working paper, City University of Hong Kong, Hong Kong SAR, China.

Liu JZ, Sockin M, Xiong W (2021) Data privacy and consumer vulnerability. Working paper, Chinese University of Hong Kong, Shenzhen, Guangdong, China.

Mikhed V, Vogan M (2018) How data breaches affect consumer credit. *J. Bank. Finance* 88:192–207.

Pentina I, Zhang L, Bata H, Chen Y (2016) Exploring privacy paradox in information-sensitive mobile app adoption: A cross-cultural comparison. *Comput. Human Behav.* 65:409–419.

Prince JT, Wallsten S (2022) How much is privacy worth around the world and across platforms? *J. Econom. Management Strategy* 31(4):841–861.

Rysman M, Schuh S (2017) New innovations in payments. *Innovation Policy Econom.* 17(1):27–48.

Schuh S, Stavins J (2016) How do speed and security influence consumers'payment behavior? *Contemporary Econom. Policy* 34(4): 595–613.

Tang H (2019) The value of privacy: Evidence from online borrowers. Working paper, The Wharton School, University of Pennsylvania, Philadelphia, PA.

Teo A-C, Tan GW-H, Ooi K-B, Hew T-S, Yew K-T (2015) The effects of convenience and speed in m-payment. *Industry Management Data Systems.* 115(2):311–331.

U.S. Government Accountability Office (2007) Personal information: Data breaches are frequent, but evidence of resulting identity theft is limited; however, the full extent is unknown. Government Report. https://www.gao.gov/products/gao-07-737.

Wottrich VM, van Reijmersdal EA, Smit EG (2018) The privacy tradeoff for mobile app downloads: The roles of app value, intrusiveness, and privacy concerns. *Decision Support Systems* 106:44–52.

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